

HackFusion

IBM BOB Hackathon

Defense & Aerospace — Mission Readiness & Predictive Maintenance Copilot

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Problem Statement

Military organisations cannot reliably determine whether aircraft, vehicles, and equipment are mission-ready. Maintenance runs on fixed calendar schedules regardless of actual component condition. HUMS (Health & Usage Monitoring System) sensor data that could predict failures weeks in advance sits unanalysed. When platforms fail unexpectedly, operational readiness drops and recovery takes weeks. The US military spends \$90B/year on maintenance — shifting to predictive approaches saves billions.



01

The Calendar Trap Challenge

Military aircraft and combat vehicles operate on rigid calendar-based maintenance rather than real-time component wear, leading to over-maintenance of healthy parts and unexpected failures in field ops.

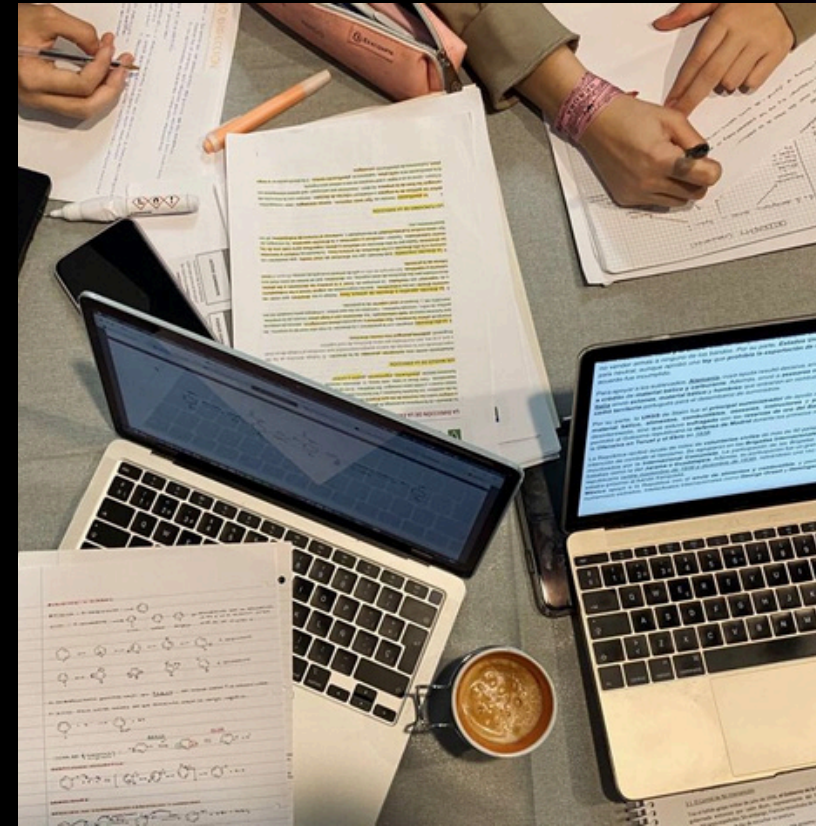
02

The Data Paradox Challenge

Gigabytes of HUMS vibration, thermal, and pressure telemetry sit unanalyzed across disparate legacy repositories.

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Proposed Solution (The Copilot Framework) :-



01

Mission Readiness Classifier

Evaluates whole-platform health against target operational mission parameters (e.g., flight duration, climate extremes, payload).

02

RAG Technical Copilot

Natural language assistant grounded in aerospace maintenance manuals Tech Orders) that diagnoses alerts and guides technicians.

03

Dynamic Maintenance Scheduler

Replaces calendar queues with a criticality-weighted priority engine balancing spare-parts inventory, technician hours, and sortie schedules.

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Predictive Modeling & Failure Forecasting (ML Layer)

Feature Engineering, Model Benchmarking, Anomaly Detection.

Our dual-tier machine learning pipeline converts raw HUMS telemetry into real-time operational decisions. By coupling unsupervised anomaly detection with supervised Remaining Useful Life (RUL) regression, the system identifies early degradation patterns weeks before mechanical failure occurs. Mathematical degradation outputs are continuously benchmarked against historical turbofan profiles to clear assets for active sorties or flag mission-critical maintenance.

01

Feature Engineering from HUMS Telemetry:

Extracts frequency-domain characteristics using Fast Fourier Transforms (FFT) alongside rolling statistical measures (RMS, peak-to-peak amplitude) to isolate harmonic vibration stress and thermal deltas.

02

Dual-Tier ML Architecture:

Employs Isolation Forests and Autoencoders for baseline anomaly detection, combined with gradient-boosted trees (XGBoost) and LSTMs to compute deterministic component RUL scores.

03

Sortie-Aware Thresholds:

Maps remaining component life directly against upcoming flight profiles, automatically triaging platforms into Mission-Ready (>100h), Scheduled Service (20–100h), or Grounded (<20h).

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Automated Work Order & Maintenance Prioritization

01

Pre-Mission Sortie Validation

Simulates asset durability against planned operational profiles (flight duration, altitude, temperature) to guarantee a $> 1.5\times$ component safety factor before clearing takeoff.

02

Technology Integration

Instantly compiles telemetry anomalies into standardized maintenance tickets, mapping detected faults to official MIL-STD defect codes and relevant subsystem tags.

03

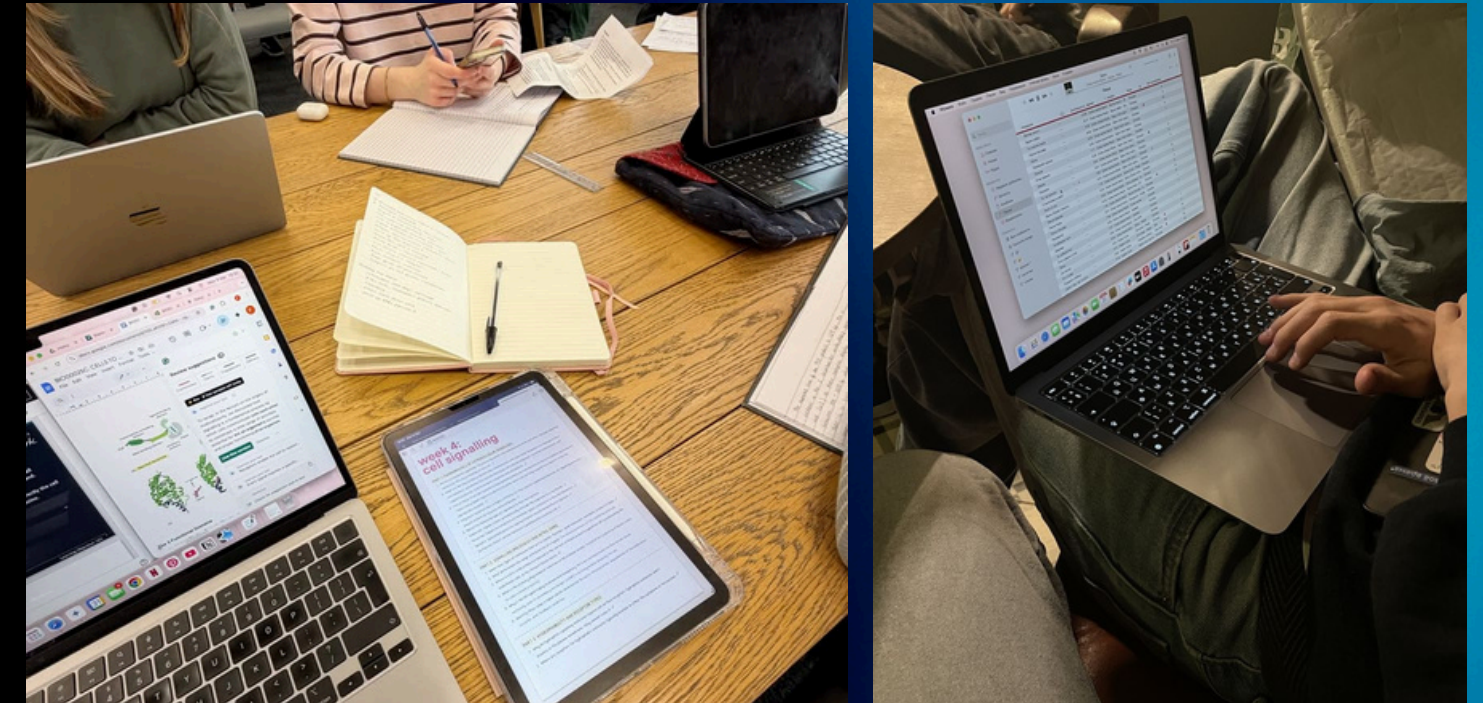
Supply Chain & Inventory Check

Cross-references predicted part replacements against depot-level inventory, verifying spare parts availability and tools before an aircraft enters the maintenance bay.

04

Guided RAG Troubleshooting

Pulls exact technical orders (TOs) and step-by-step teardown checklists directly into the technician's view, reducing diagnostic and turnaround times by over 40%.



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Feasibility, Hackathon Execution & Datasets

To prove real-world viability within the hackathon timeframe, our team built an operational end-to-end prototype grounded in publicly verified aerospace telemetry benchmarks. Rather than using mock numbers, the engine runs live inference on multi-cycle turbofan run-to-failure datasets, translating high-frequency vibration and thermal sensor streams into actionable remaining flight hours and instant maintenance work orders.



01

NASA C-MAPSS Benchmark Validation

Leverages gold-standard turbofan run-to-failure datasets to train and evaluate our RUL regression models across diverse operating regimes, thermal stresses, and degradation conditions.

02

Live Inference & Demo Flow

Demonstrates an end-to-end operational pipeline: flags degraded airframe telemetry, predicts imminent component failure windows, and auto-generates actionable, prioritized technician work orders.

Real-World Impact, Scalability & Future Roadmap

Deploying predictive maintenance shifts defense operations from costly scheduled overhauls to dynamic, condition-based readiness. By eliminating unnecessary component teardowns and intercepting mechanical faults weeks in advance, the platform maximizes mission availability, cuts depot-level downtime, and directly addresses the multi-billion-dollar operational drain across defense and commercial aviation fleets.



01

Strategic & Economic ROI

Reduces unscheduled fleet downtime by an estimated 25–35% while slashing premature part replacements, directly mitigating billions in reactive logistics and emergency procurement costs.

02

Deployment Roadmap

Expands from our cloud prototype to air-gapped, edge-deployed inference units on flight-line tablets, integrating directly into military inventory databases and commercial ERP systems (e.g., IBM Maximo, SAP).

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Thank you!